

CS549 Midterm

I. QUESTION: HMM

Let us say we have 4 states:

S_1 : A, S_2 : C, S_3 : G, S_4 : T

with initial probabilities: $\Pi = [0.3, 0.4, 0.1, 0.2]$

and the transition matrix is:

$$\mathbf{A} = \begin{vmatrix} 0.2 & 0.2 & 0.3 & 0.3 \\ 0.1 & 0.3 & 0.2 & 0.4 \\ 0.4 & 0.4 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.7 \end{vmatrix}$$

- Give a sequence S consisting of 7 symbols, where each state occurs at least once. Calculate the probability of the sequence S .
- Give 10 different sequences $S_1, S_2, \dots, S_{10}$ of size 6, where each state occurs in each sequence at least once. Given those ten sequences estimate the initial probabilities for Π and $\mathbf{A}$.

II. QUESTION: LOGISTIC REGRESSION

Derive the update equation for 2-class logistic regression with L_2 regularization. L_2 regularization objective function is:

$$\log p(y|x) - \alpha \sum_{j=1}^n w_j^2$$

, and

$$\log p(y|x) = y \log \hat{y} + (1 - y) \log(1 - \hat{y})$$

, where

- n is the number of weights
- α is the regularization parameter
- x is the input of instance
- y is the real output of instance
- $\hat{y}$ is the calculated output of instance

III. QUESTION: NGRAM

Given the following sentences:

```
<s> ali topu at . </s>
<s> ali ayşeye çiçek al . </s>
<s> ali topu sev . </s>
<s> ayşe topu at . </s>
<s> ali ayşeye topu at . </s>
```

- Calculate the following unigram probabilities: $P(ali)$, $P(< s >)$, $P(ayşeye)$, $P(at)$. Assume that add- k smoothing is applied with $k = 1$.
- Calculate the following bigram probabilities: $P(ali | < s >)$, $P(topu | ali)$, $P(< /s > | .)$, $P(ayşeye | ali)$. Assume that add- k smoothing is applied with $k = 2$.

IV. QUESTION: MORPHOLOGICAL ANALYSIS

Write a pseudocode to get the possible roots of a word from the Trie data structure.

```
class TrieNode
    children : Map<char, TrieNode>
    words : Set<word>

class Trie
    root : TrieNode
```

V. QUESTION: SENTIMENT ANALYSIS

Given the following sentences, positive and negative categories; calculate the category for the given test sentences. Use Laplace smoothing to calculate probabilities.

Sentence	Category
hep tekrar hep tekrar	-
süper kötü bir film	-
çok iyi film	+
karakter gelişimi iyi	+
yazın en iyi film bu	+
çok süper film	?
iyi karakter çok kötü film	?

VI. QUESTION: MORPHOLOGICAL DISAMBIGUATION

In Turkish morphological disambiguation, there are two major problems. The first problem is to identify the correct root forms, and the second problem is to identify the correct morphemes. Assume that we are only interested in the first problem, and assume also that you have a morphologically disambiguated corpora. Propose a root form disambiguation algorithm to solve that problem. Use only algorithms until the end of the first midterm.